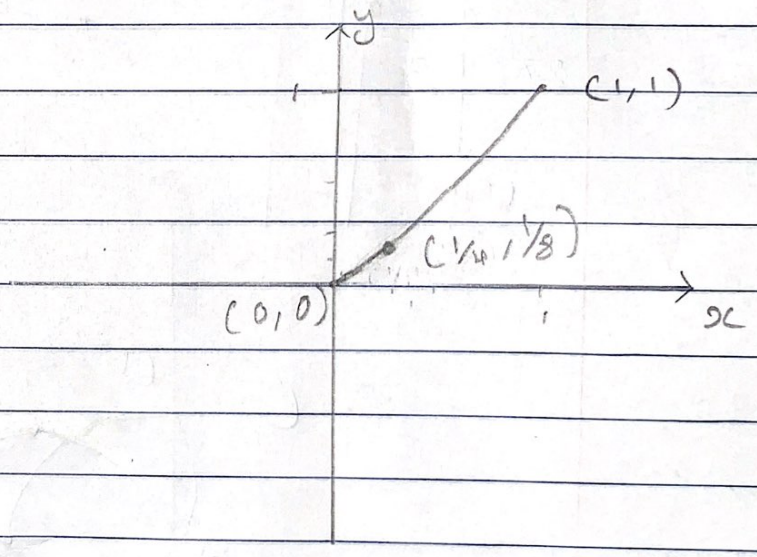
Quiz 19

(C16) $x = t^2$, $y = t^3$ $\quad 0 \le t \le 1$

t	x	y
0	0	0
1/2	1/4	1/8
1	1	1



(C18) $r = 3\cos\theta$ and $r = 1 + \cos\theta$

$\Rightarrow \quad 1 + \cos\theta = 3\cos\theta$

$\Rightarrow \quad 1 = 3\cos\theta - \cos\theta$

$\Rightarrow \quad \cos\theta\,(3-1) = 1$

$\Rightarrow \quad \cos\theta\,(2) = 1$

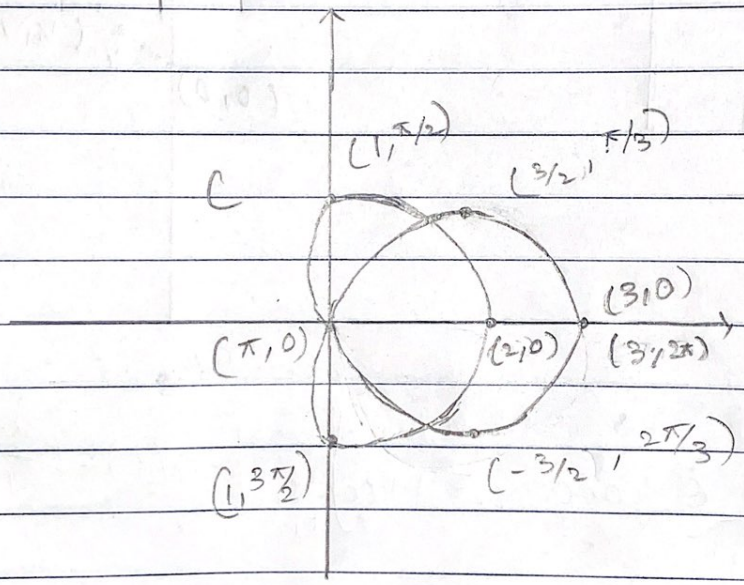
$\Rightarrow \quad \cos\theta \in 1/2 \quad \Rightarrow \quad \theta = \pi/3$

$r = a\cos\theta \quad \Rightarrow r = 3\cos\theta$

$r = a + b\cos\theta \quad \Rightarrow r = 1 + \cos\theta$

θ	r
0	3
$\pi/3$	$3/2$
$2\pi/3$	$-3/2$
2π	3

θ	r
0	2
$\pi/2$	1
$3\pi/2$	1
π	0
2π	2



(C19) $$A = \int_0^{\pi/3} \left((3\cos\theta)^2 - (1+\cos\theta)^2 \right) d\theta$$

$$A = \int_0^{\pi/3} \left(9\cos^2\theta - 1 + 2\cos\theta - \cos^2\theta \right) d\theta$$

$$A = \int_0^{\pi/3} (8\cos^2\theta - 1 - 2\cos\theta)\, d\theta$$

$$A = \int_0^{\pi/3} 8\cos^2\theta\, d\theta - \int_0^{\pi/3} d\theta - \int_0^{\pi/3} 2\cos\theta\, d\theta$$

$$A = 4\int_0^{\pi/3} (1+\cos(2\theta))\, d\theta - \pi/3 - \sqrt{3}$$

$$A = 4\left(\pi/3 + \left(\frac{\sqrt{3}}{4}\right)\right) - \pi/3 - \sqrt{3}$$

$$A = 4\pi/3 + \sqrt{3} - \sqrt{3} - \pi/3$$

$$A = 3\pi/3 = \pi$$

$$\boxed{A = \pi}$$